

Engineering Statistics

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Chapter 1: Introduction

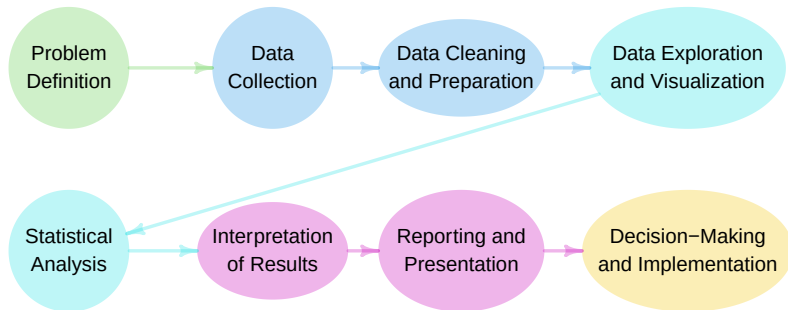
Overview

- ▶ **Definition:** What is statistics?
- ▶ **The Data Analysis Workflow:** From raw data collection to reproducible reporting.
- ▶ **Data Type:** Distinguishing between categorical and quantitative variables.
- ▶ **Statistical Toolkits:** Overview of the mathematical and software tools used in statistics.
- ▶ **Introduction to R, RStudio, and Gemini:** Setting up your environment for computing.

Statistics

- ▶ Statistics is the discipline that concerns the collection, organization, analysis, interpretation, and presentation of data.
- ▶ Collection and organization: how do you get the data? Design of experiments, Survey sampling, Data cleaning, table/matrix, etc.
- ▶ Analysis: how do you use the data? Probability tools, regressions, parameter estimations, hypothesis testing, prediction, statistical quality control, and reliability analysis, etc.
- ▶ Interpretation and presentation: what do you find from the data? Data visualization/making plots, data summarization, making conclusions, interpreting in the context of the problem, etc.

Basic Workflow of Data Analysis: Illustration



Basic Workflow of Data Analysis: Eight Steps

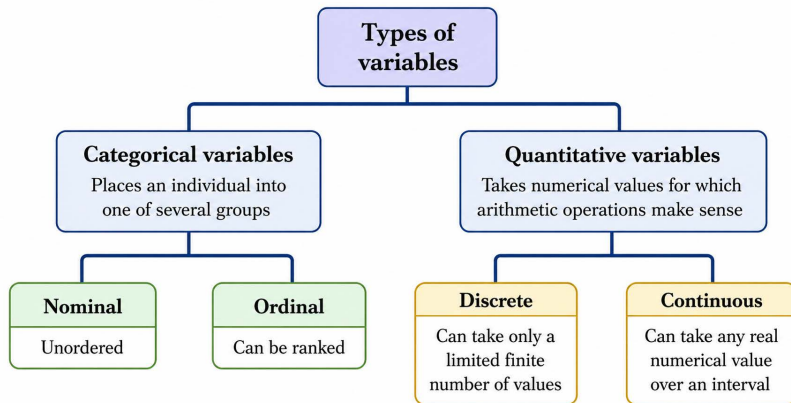
1. Problem definition: Clearly identify the question or objective of the study.
2. Data collection: Obtain relevant data from experiments, surveys, sensors, or databases.
3. Data cleaning and preparation: Remove errors and organize data into an analyzable form.
4. Data exploration and visualization: Examine the data patterns using plots and summaries.
5. Statistical analysis or modeling: Apply statistical methods or models to extract insights.
6. Interpretation of results: Translate findings into meaningful conclusions.
7. Reporting and presentation: Communicate results through reports, slides, or dashboards.
8. Decision-making and implementation: Use the conclusions to guide actions or policies.

Individuals and variables

- ▶ Individuals are the objects described by a set of data.
- ▶ Examples: people, animals, aircraft, machine. etc.
- ▶ A variable is any characteristic of an individual or object.
- ▶ Examples: fish size, animal age, cause of death, time, speed, temperature, etc.

Types of variables

- Determining the type of the variable is simple but very important in analyzing data numerically and graphically.



The Mathematical Foundations of Statistics

► **Probability Theory**

The language of uncertainty. It provides the framework for modeling random variables and the theoretical distributions (Normal, Binomial, Poisson) that data follow.

► **Linear Algebra**

The engine for multidimensional data. Used for representing datasets as matrices, solving OLS regressions.

► **Calculus**

The tool for optimization.

- *Differential*: Used to find Maximum Likelihood Estimators (MLE).
- *Integral*: Used to calculate areas under density curves to find probabilities.

Main Models in Statistical Inference

► Regression Models

- **Linear Regression:** Models continuous outcomes as a linear combination of predictors.
- **Generalized Linear Models (GLM):** Extends regression to non-normal data (e.g., Logistic for binary data, Poisson for counts).

► Comparison Models (Group Differences)

- **t-Tests / ANOVA:** Infers whether differences between group means are statistically significant.
- **Chi-Square:** Tests for relationships between categorical variables.

► Estimation Frameworks

- **Frequentist (Likelihood-based):** Uses Maximum Likelihood Estimation (MLE) to find parameters that make the observed data most probable.
- **Bayesian Models:** Incorporates prior knowledge and updates it with data to produce a *posterior* probability distribution.

Predictions

- ▶ **Statistical Prediction**

- ▶ **Time Series:** Using *ARIMA* or *Exponential Smoothing* to predict future values based on past trends and seasonality.
- ▶ **Gaussian Processes (GP):** A powerful approach that provides a full probability distribution over possible functions, offering built-in uncertainty quantification.

- ▶ **Algorithmic Prediction (Machine Learning)**

- ▶ **Tree-Based Models:** Random Forests and Gradient Boosting (XGBoost) for handling non-linear relationships.
- ▶ **Regularization:** *LASSO* and *Ridge* regression to prevent overfitting and improve out-of-sample accuracy.

- ▶ **The Modern Frontier**

- ▶ **Neural Networks:** Deep learning for high-dimensional prediction (image, speech, and text).
- ▶ **Generative Models:** Using tools like Gemini to simulate data scenarios and synthetic populations.

Computing Tools: Introduction to R

- ▶ R is a programming language and software environment for statistical computing and graphics.
- ▶ R is open source.
- ▶ Widely used in statistics, data science, and research
- ▶ Extensive package ecosystem via CRAN
- ▶ Strong visualization capabilities (e.g., `ggplot2`)

Introduction to RStudio

- ▶ RStudio is an integrated development environment (IDE) for R that makes it easier to write code, run analyses, and manage projects.
- ▶ Provides a source editor, console, environment pane, and file browser
- ▶ Integrates with R for data analysis, graphics, and reporting
- ▶ Supports R Markdown for dynamic documents and reproducible reports
- ▶ Offers project management tools for organizing files and workflows
- ▶ Available on Windows, macOS, and Linux (and as RStudio Server)

Example 1: Analyzing Security Data

Scenario: Measuring the "Time to Patch" (in hours) for 10 vulnerabilities.

R Code: Vectors and Basic Stats

```
# 1. Create a numeric vector
ttp <- c(33, 30, 29, 27, 25, 26, 29, 36, 30, 28)
```

```
# 2. Calculate summary statistics
mean(ttp)      # Returns 29.3
median(ttp)    # Returns 29
sd(ttp)        # Standard Deviation: 3.16
```

```
# 3. Logical Subsetting (Find slow patches)
slow_patches <- ttp[ttp > 30]
```

- ▶ `c()` combines individual values into a single **vector**.
- ▶ R functions are **vectorized**, meaning they operate on the whole list at once.

Example 2: Process Monitoring with Data Frames

Scenario: Tracking reactor temperature and catalyst type.

R Code: Data Frames and Plots

```
# 1. Build a Data Frame
reactor_data <- data.frame(
  temp = c(150, 152, 149, 155, 151),
  catalyst = c("Zeolite", "Alumina", "Alumina",
    "Pt", "Zeolite")
)

# 2. Accessing Data
reactor_data$temp # Access one column

# 3. Simple Visualization
hist(reactor_data$temp, col="lightblue",
main="Reactor Temp Dist")
```

- ▶ **Data Frames** allow mixed types (numeric and categorical).
- ▶ The `$` operator is the shortcut to select a specific variable.

New Tools: GenAI and Agentic AI

- ▶ **Generative AI (GenAI):** AI that focuses on **creating new content**.
 - ▶ **Multimodal:** Generates text, images, audio, video, and code.
 - ▶ **Interactive:** Responds to prompts and assists with problem solving.
- ▶ **Agentic AI:** AI that can **plan and take actions** to accomplish a goal.
 - ▶ **Autonomous:** Breaks a goal into multiple steps and decides what to do next.
 - ▶ **Tool-using:** Can search the web, analyze data, run code, work with files, and interact with other tools.
 - ▶ **Iterative:** Evaluates intermediate results and adjusts its approach when needed.

Google Gemini

- ▶ Google's family of **multimodal, reasoning, and agentic AI** models.
- ▶ **Gemini 3.5/3.6:** New-generation models with improved reasoning, coding, multimodal understanding, and tool use.
- ▶ **Deep Think:** Advanced reasoning for difficult problems in mathematics, science, and engineering.
- ▶ **Multimodal Generation:** Works across text, images, audio, video, and code.
- ▶ **Agentic AI:** Can plan and carry out multi-step tasks using tools, including web search, code execution, files, and computer use.
- ▶ **Google Ecosystem:** Integrated with Google Search, Gmail, Docs, Drive, and other Google services.

Coding with Gemini: Generating R Code

The Prompting Strategy

- ▶ **Be Specific:** Define your data structure, variables, and library preferences.
- ▶ **Role Play:** Ask Gemini to act as a role such as "Patient Coding Tutor", "Statistical Consultant", "Senior R Developer."
- ▶ **Iterative Refinement:** Start with a base model, then ask for visualizations or diagnostics.
- ▶ **Debugging:** Paste an error message to find the fix.

Example Prompt

"Write R code using ggplot2 to create a scatter plot of 'age' vs 'income' from a dataframe 'df', and add a linear regression line with 95% confidence intervals."

Trend for Tools Used in Statistical Analysis



- ▶ The tools used in statistical analysis have always been evolving.
- ▶ **Mechanical calculators:** Supported early statistical calculations through manual operation.
- ▶ **Mainframe computers:** Enabled large-scale and more complex data analysis.
- ▶ **Statistical software:** Made modeling, visualization, and inference broadly accessible.
- ▶ **Artificial intelligence:** Supports automated coding, interpretation, and data-driven decision making.